Personal Desktop Agent

System Study Guide

Multimodal accessibility desktop control · iPad sensor hub + RTX 5090 inference

Architecture · 4-gate routing · 12 state machines · 42-table data model · 2026-06-13

Agenda

1 **Project Overview**

What it is, the end-to-end pipeline, 4-gate routing, models, the agent kernel

2 **State Machines**

12 machines — iPad sensors/UI (5) + PC agent kernel (7)

3 **Data Model**

Two-tier storage + 42-table agent.db grouped into 5 ER views

SECTION 1

Project Overview

The mission, the pipeline, and how a command is routed

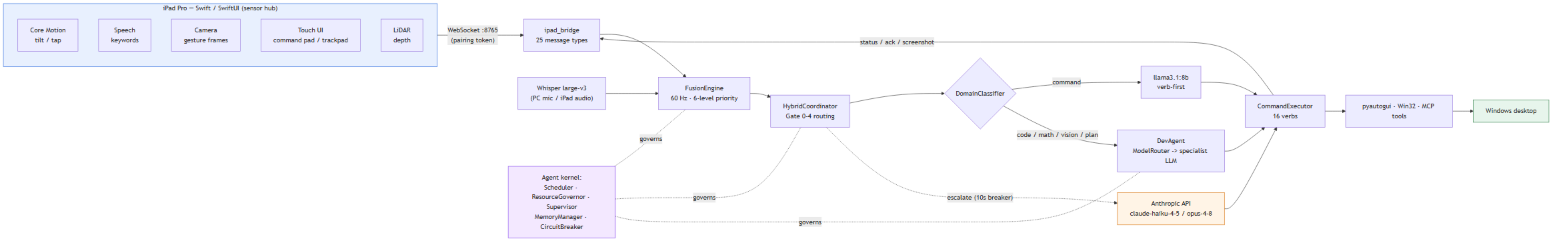
What Is This?

Overview

- **Multimodal accessibility desktop control for a single user with rheumatoid arthritis**
  - An iPad Pro is the sensor hub + primary touch surface; a Windows PC + RTX 5090 runs inference and executes desktop actions
- **Four control surfaces, one vocabulary**
  - Voice · hand gesture · iPad tilt · direct touch → a 16-verb action vocabulary
- **Local-first, privacy-first**
  - 100% local inference (Ollama → vLLM); cloud (Anthropic API) only as an escalation fallback
  - Gate 0 forces sensitive input local; every tool call lands in an append-only, hash-chained audit log
- **Scale of the system today**
  - 16 action verbs · 6-level sensor fusion @ 60 Hz · 42-table operational DB · ~1,440 tests

End-to-End Architecture

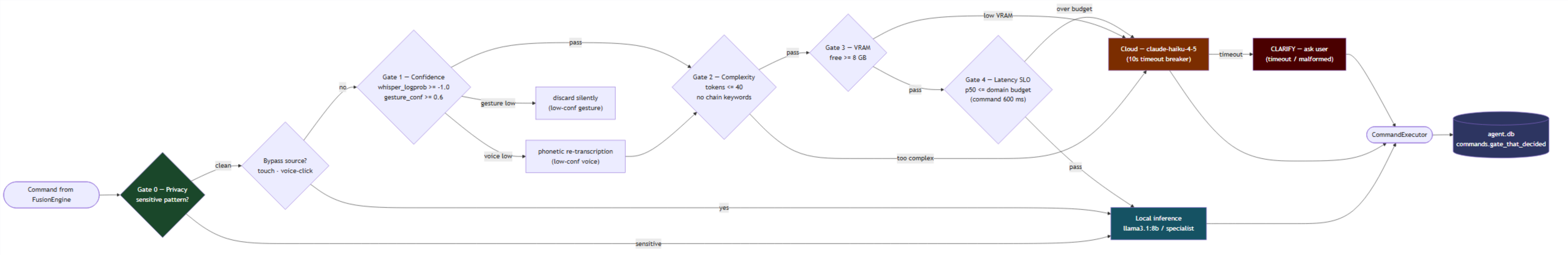
iPad sensors stream over WebSocket → fused at 60 Hz → routed → executed on the Windows desktop



How a Command Is Routed — 4 Gates

Overview

Each command passes privacy → confidence → complexity → VRAM → latency gates; the deciding gate is logged



Inference Tiers

Overview

Domain	Model	Notes
Command	llama3.1:8b (4.6 GB)	verb-first; ~190 ms warm p50
Code / Plan	qwen3-coder:30b	thinking ON; DevAgent specialist
Math	deepseek-r1:8b	chain-of-thought retained
Vision	qwen3-vl:30b	UI target → pixel grounding
General	gemma3:27b / gemma4:12b	resident slot, no eviction churn
Cloud fallback	claude-haiku-4-5 / opus-4-8	command / dev; 10 s breaker

Action Vocabulary — 16 Verbs

Overview

11 ACCESSIBILITY VERBS (iPad sensor pipeline)

- CLICK · MOUSEDOWN · MOUSEUP
- SCROLL · TYPE · DICTATE
- OPEN · CLOSE · HOTKEY
- CLARIFY · SCREENSHOT
  - Verb-first format from llama3.1:8b

5 DEV-AGENT VERBS (specialist models)

- WRITE_FILE · RUN_TERMINAL
- EXPLAIN · SEARCH_WEB
- READ_SCREEN
  - Free-form; emitted via DevAgent
  - DomainClassifier picks the pipeline; CommandExecutor handles all 16

The Agent Kernel — AIOS Primitives

Overview

- AccessibilityScheduler — priority queue, 5 tiers; accessibility/voice/gesture run concurrently, dev/background are semaphore-gated so they never starve the user
- ResourceGovernor — pain-aware; on a flare it evicts heavy models, pauses the indexer + dev admission, and relaxes sensor thresholds (hysteresis 0.6 / 0.4)
- Supervisor — one-for-one liveness watchdog; restarts dead loops under a bounded budget, latches FAILED and degrades gracefully (TTS warning)
- CircuitBreaker — latches a down inference backend so it fast-fails instead of costing a full timeout per call
- MemoryManager — a syscall façade over agent.db + SemanticMemory with schema-validated writes
- Saga + goal_queue — durable, idempotent, rollback-safe autonomous execution

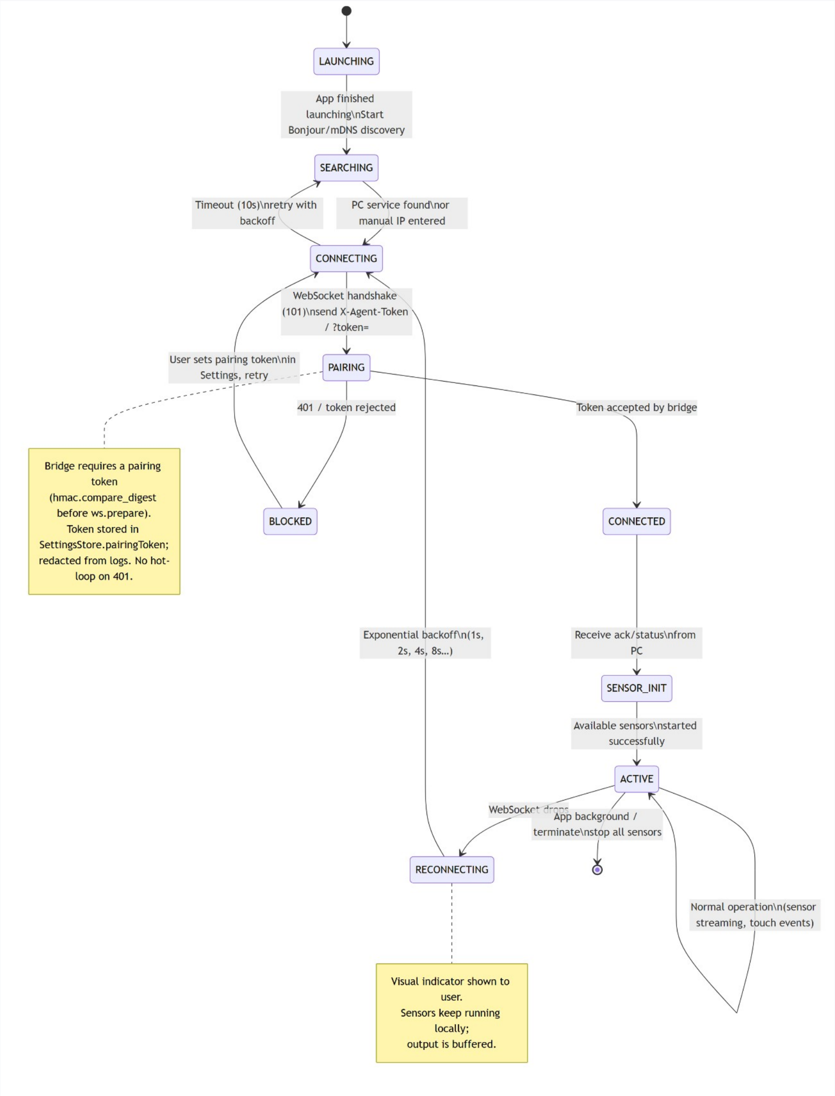
→ Each of these is a state machine. The next section diagrams them.

SECTION 2

State Machines

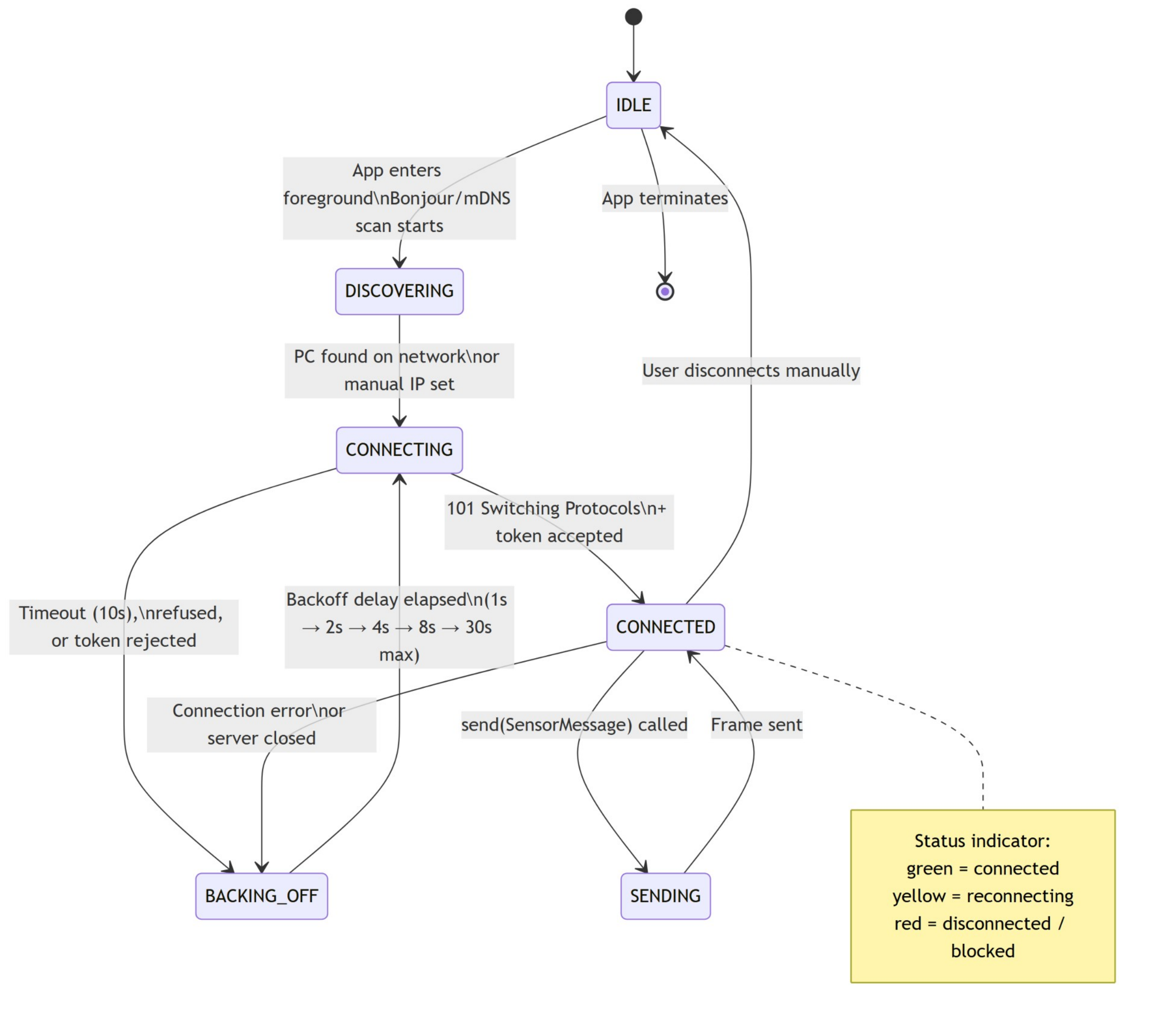
12 machines — how each subsystem behaves over time

App Top-Level Lifecycle

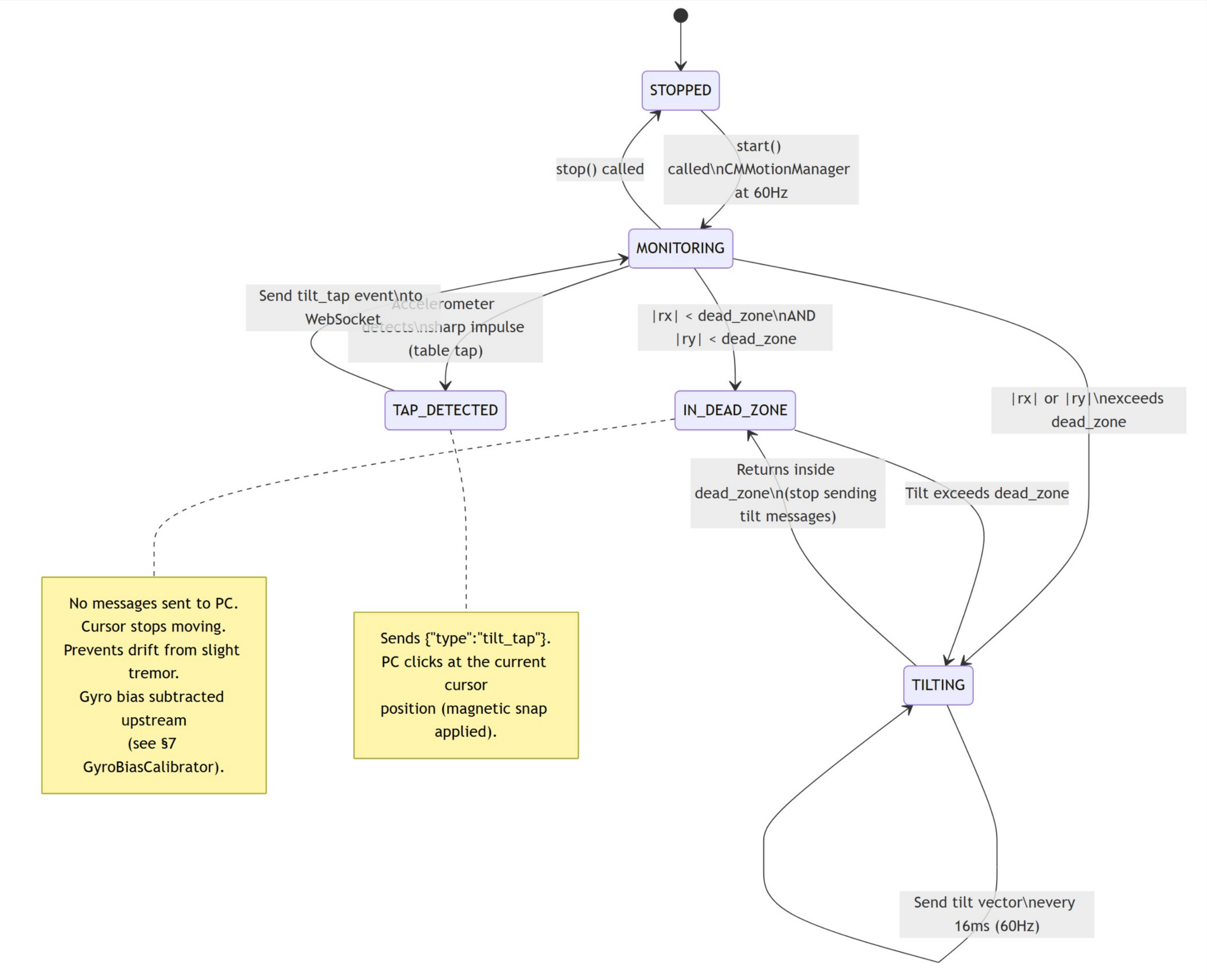


Source: DesktopAgentApp · SensorManager

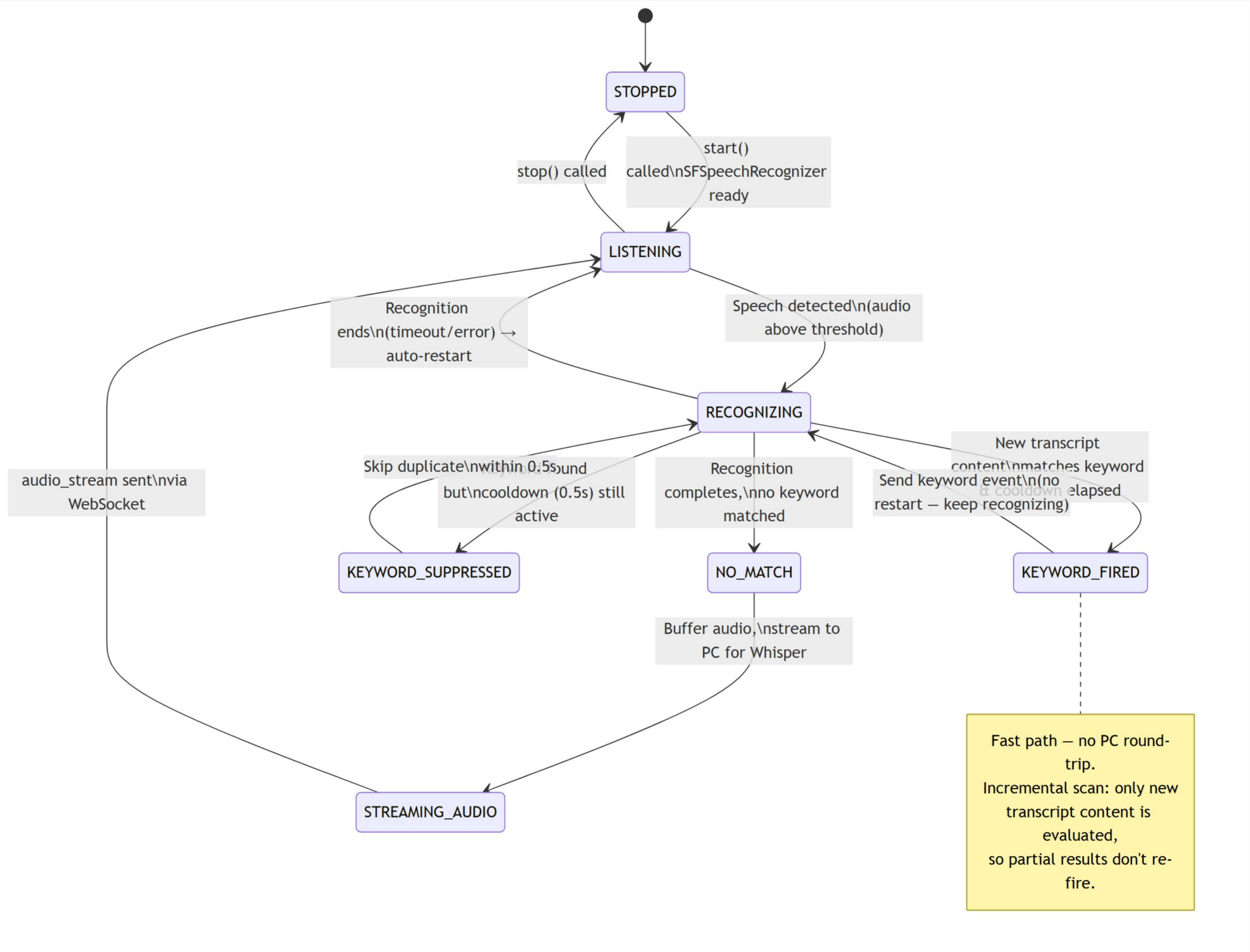
WebSocket Connection



Tilt Navigation

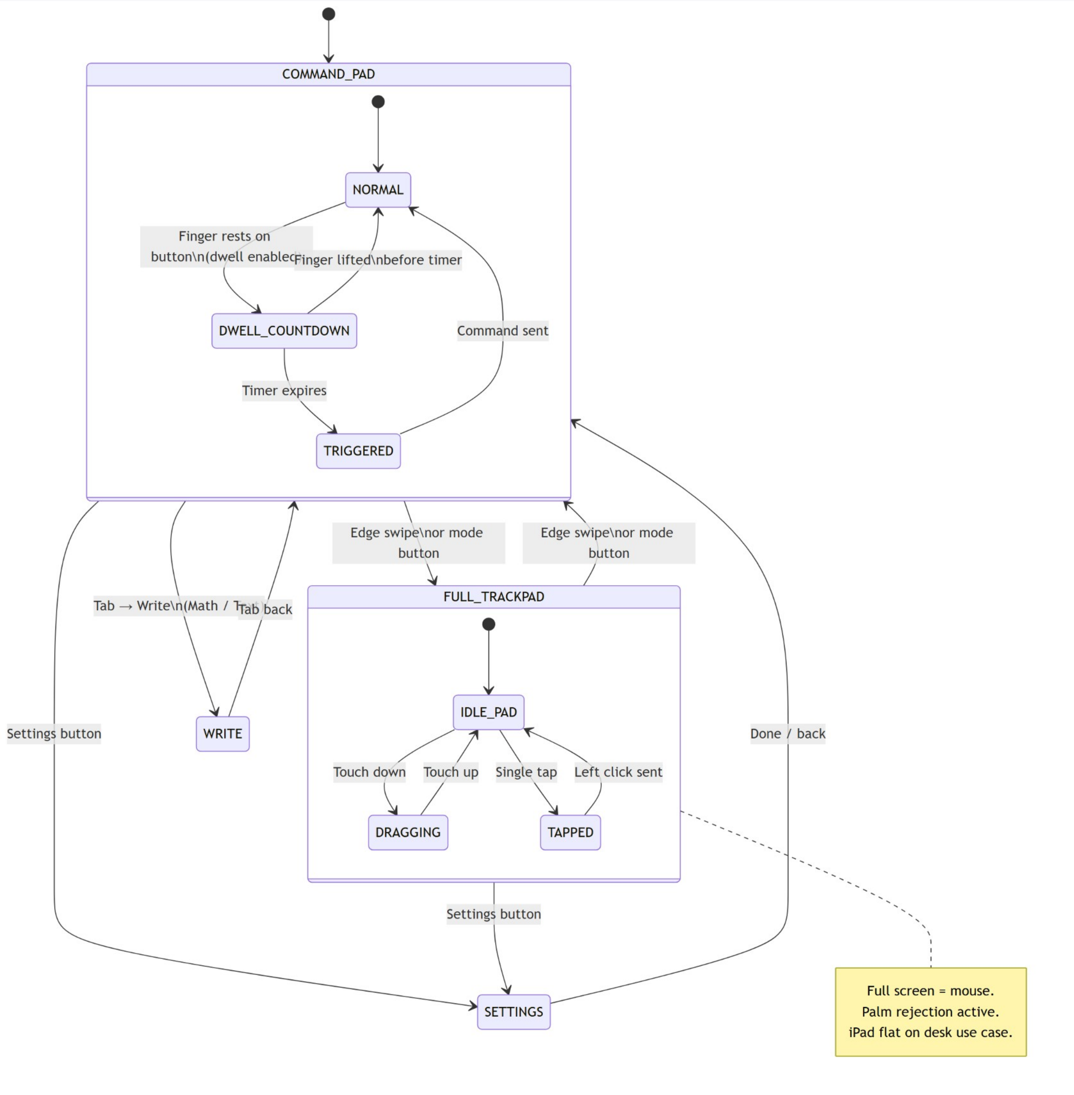


Keyword Listener



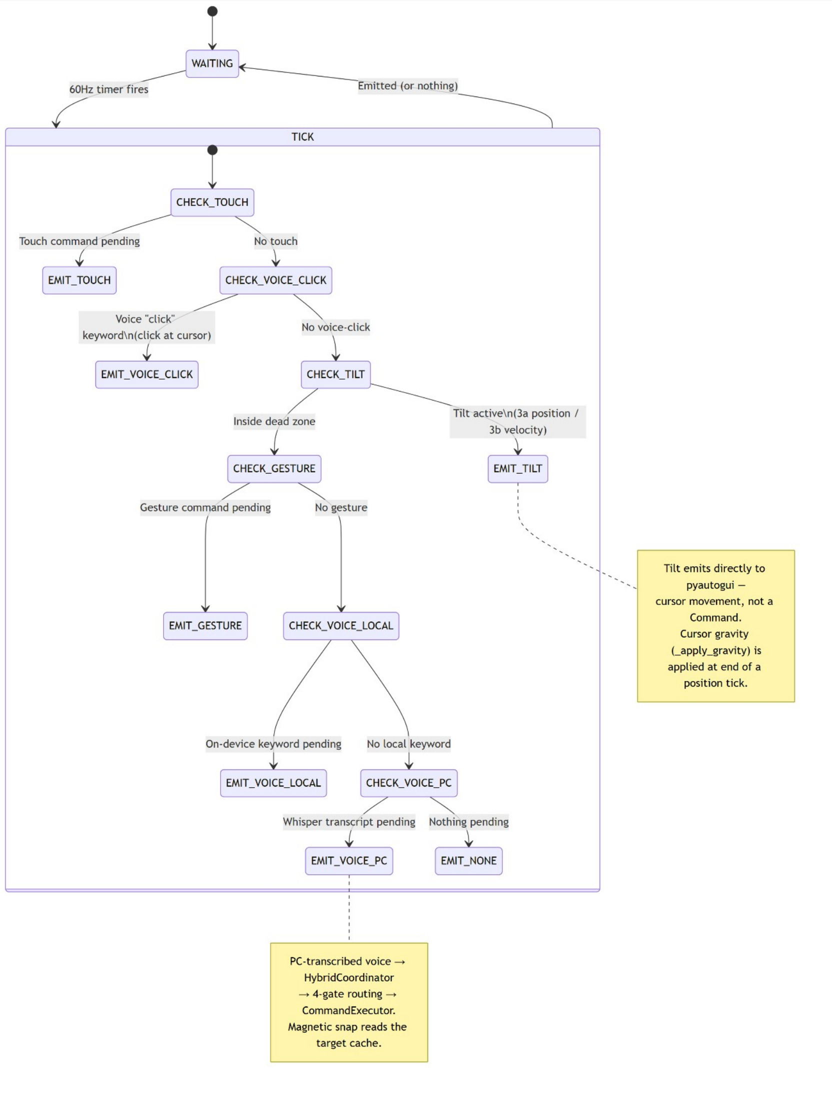
Source: KeywordListener.swift (Speech)

UI View Mode

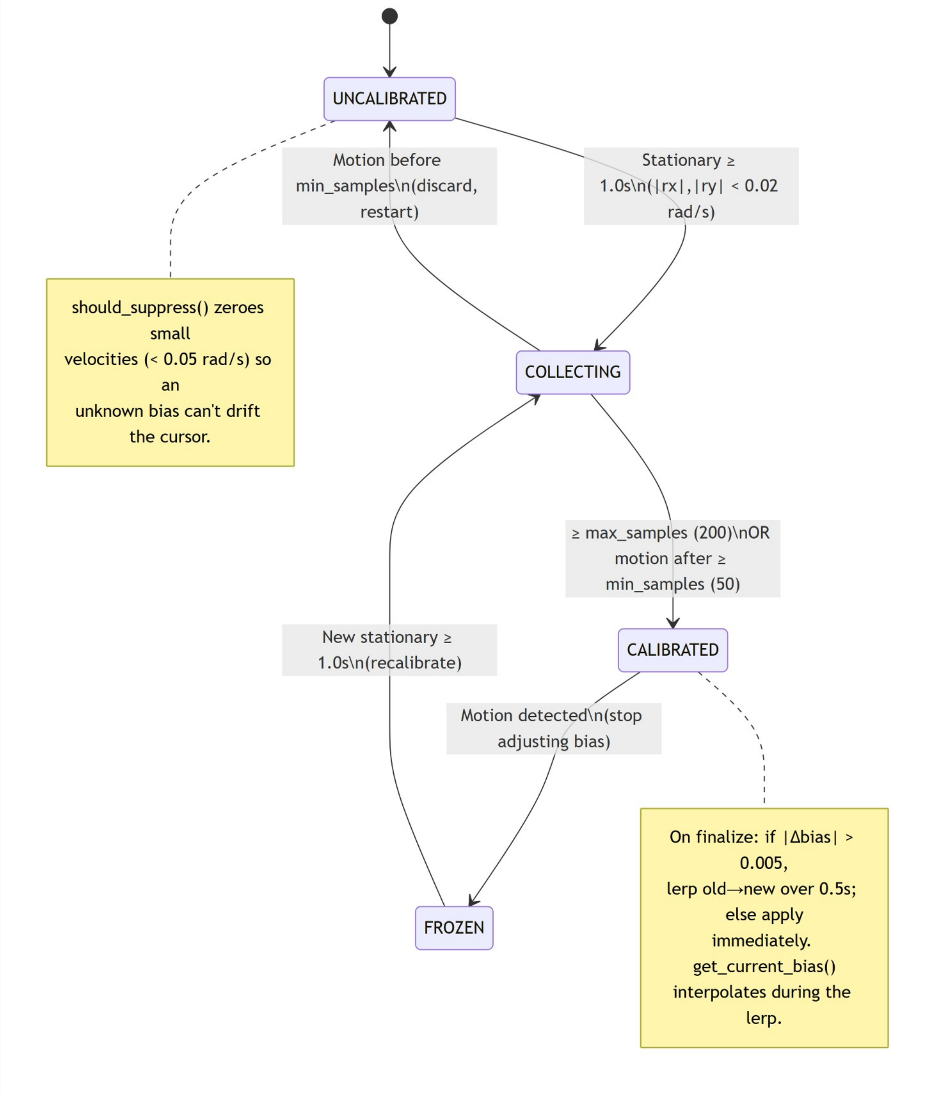


Source: SwiftUI views

FusionEngine — 6-Level Tick

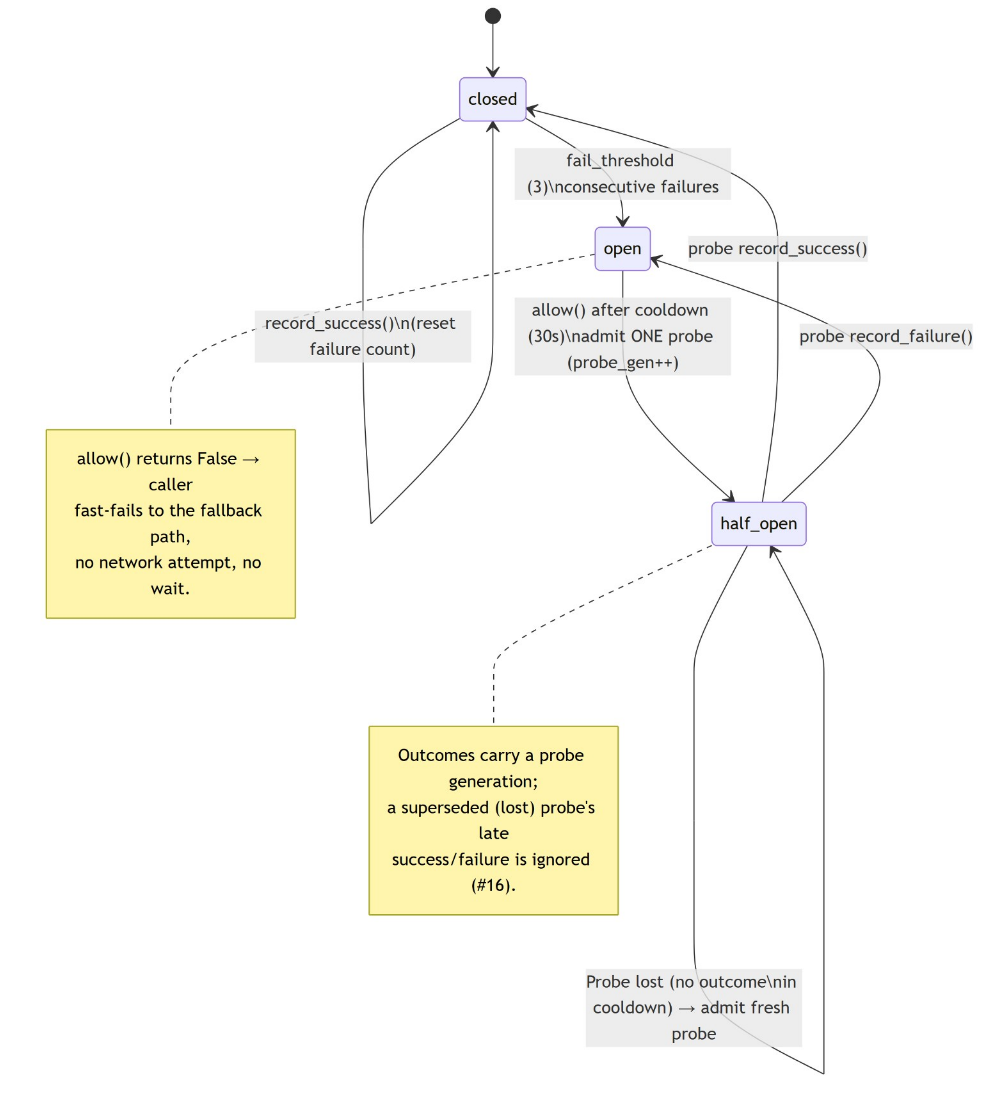


Gyro Bias Calibrator

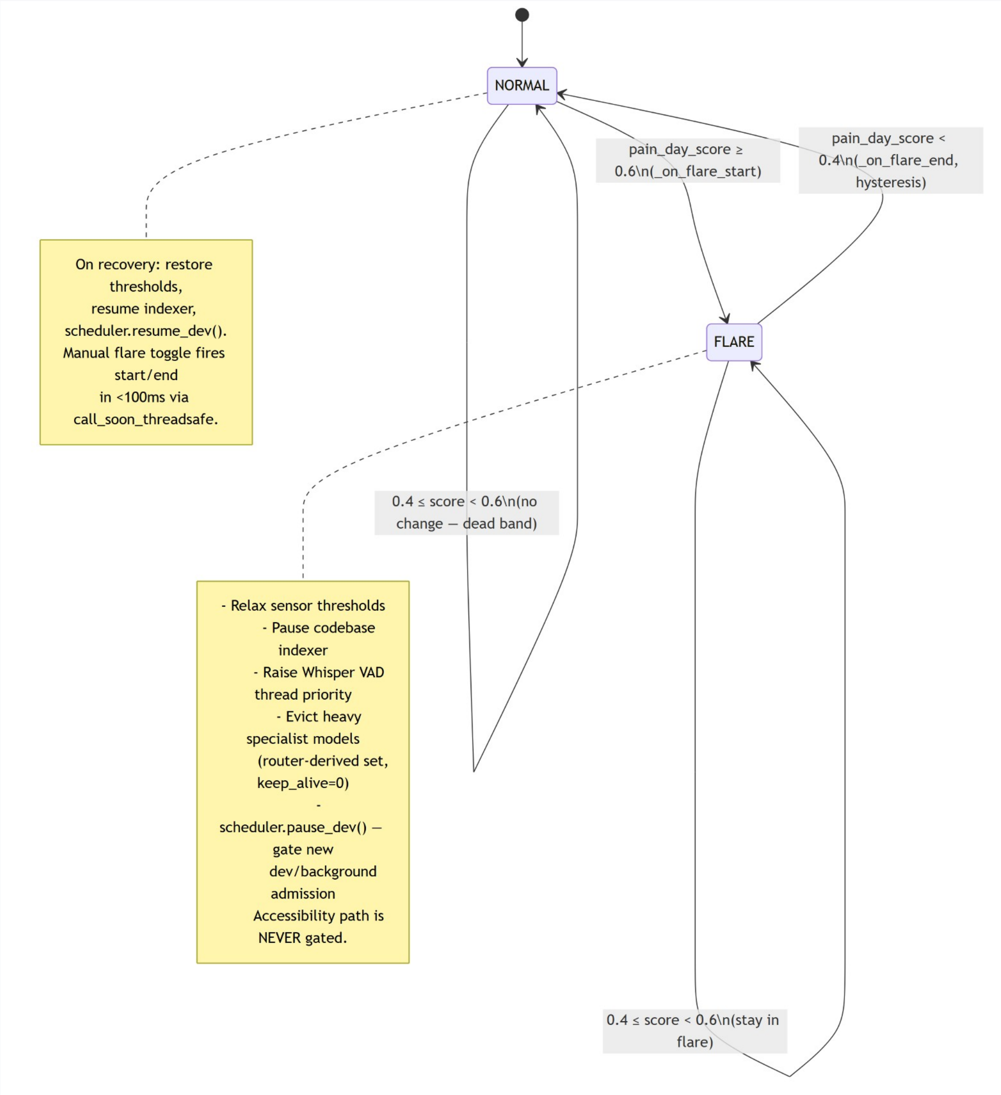


Source: calibration/gyro_bias_calibrator.py

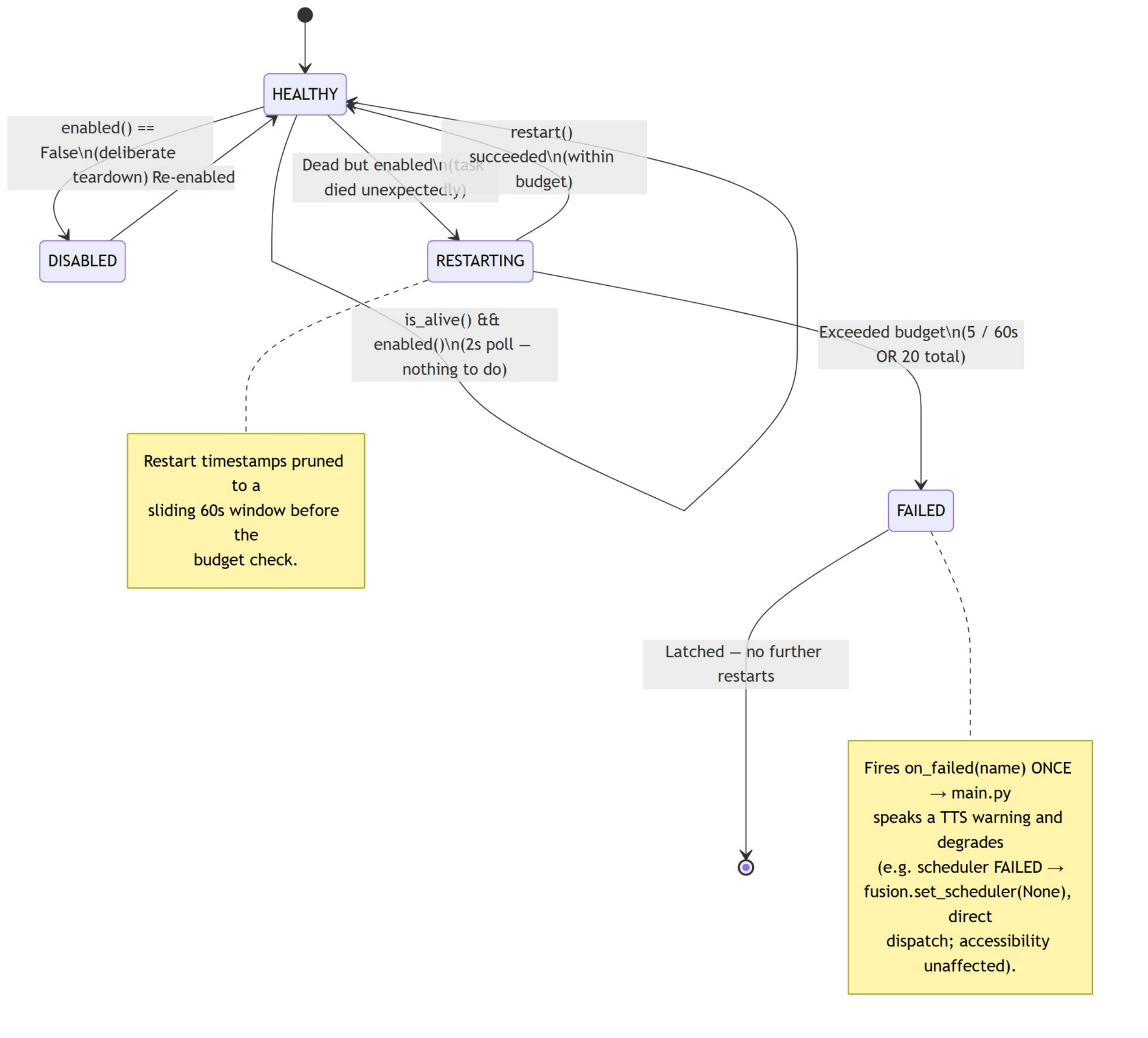
Circuit Breaker



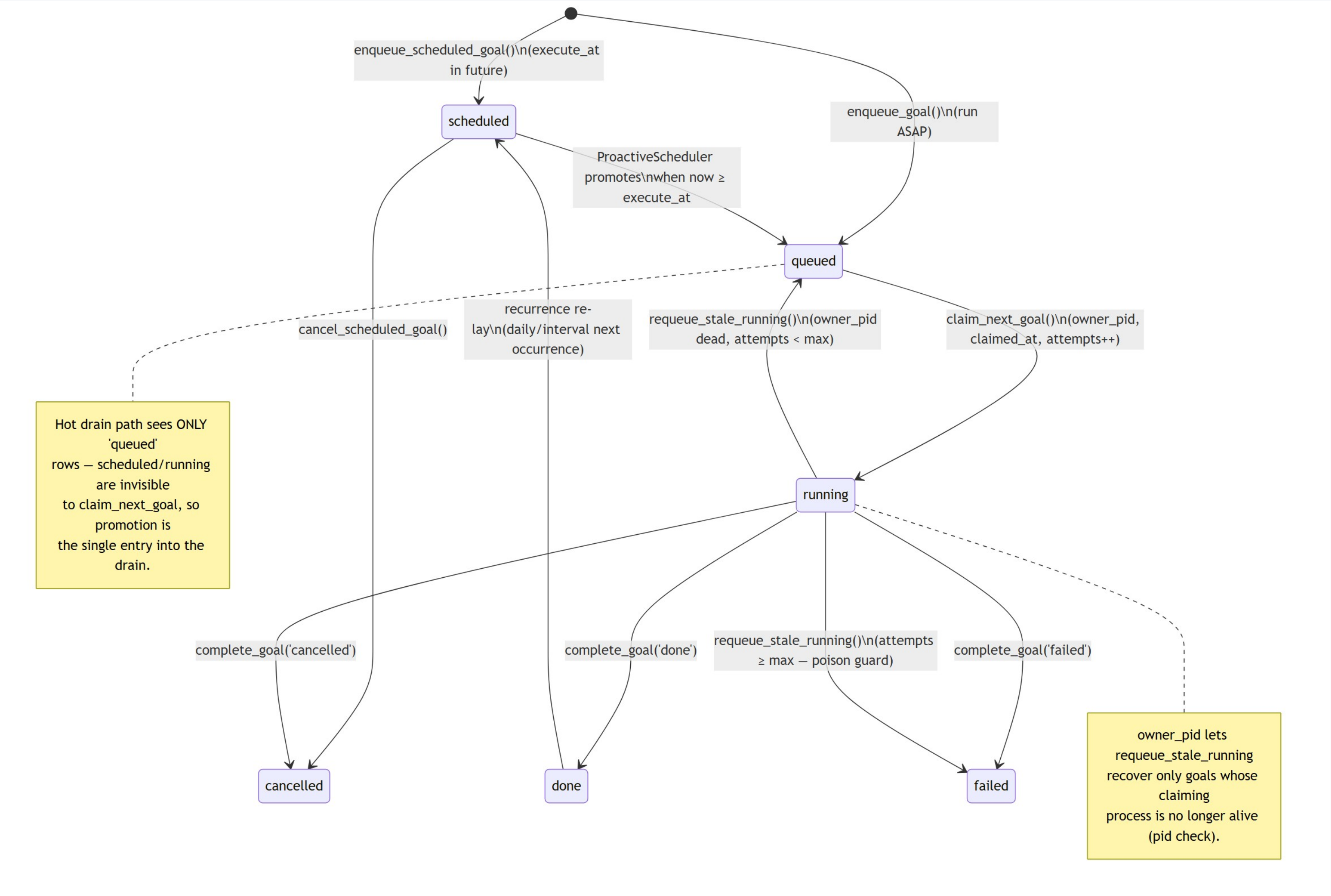
Resource Governor (Flare)



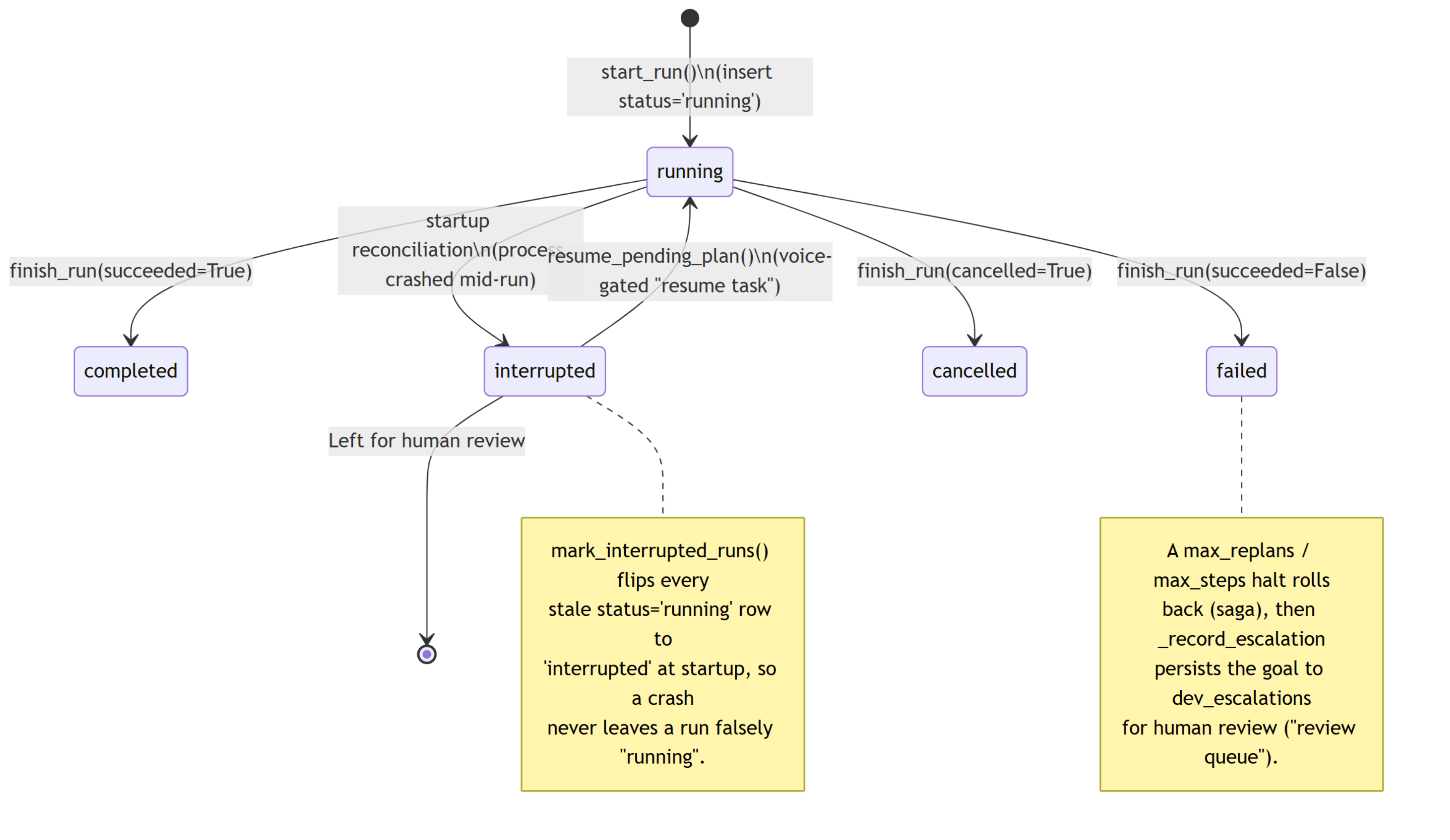
Supervisor Restart Policy



Goal-Queue Row Lifecycle



Agent-Run Lifecycle



SECTION 3

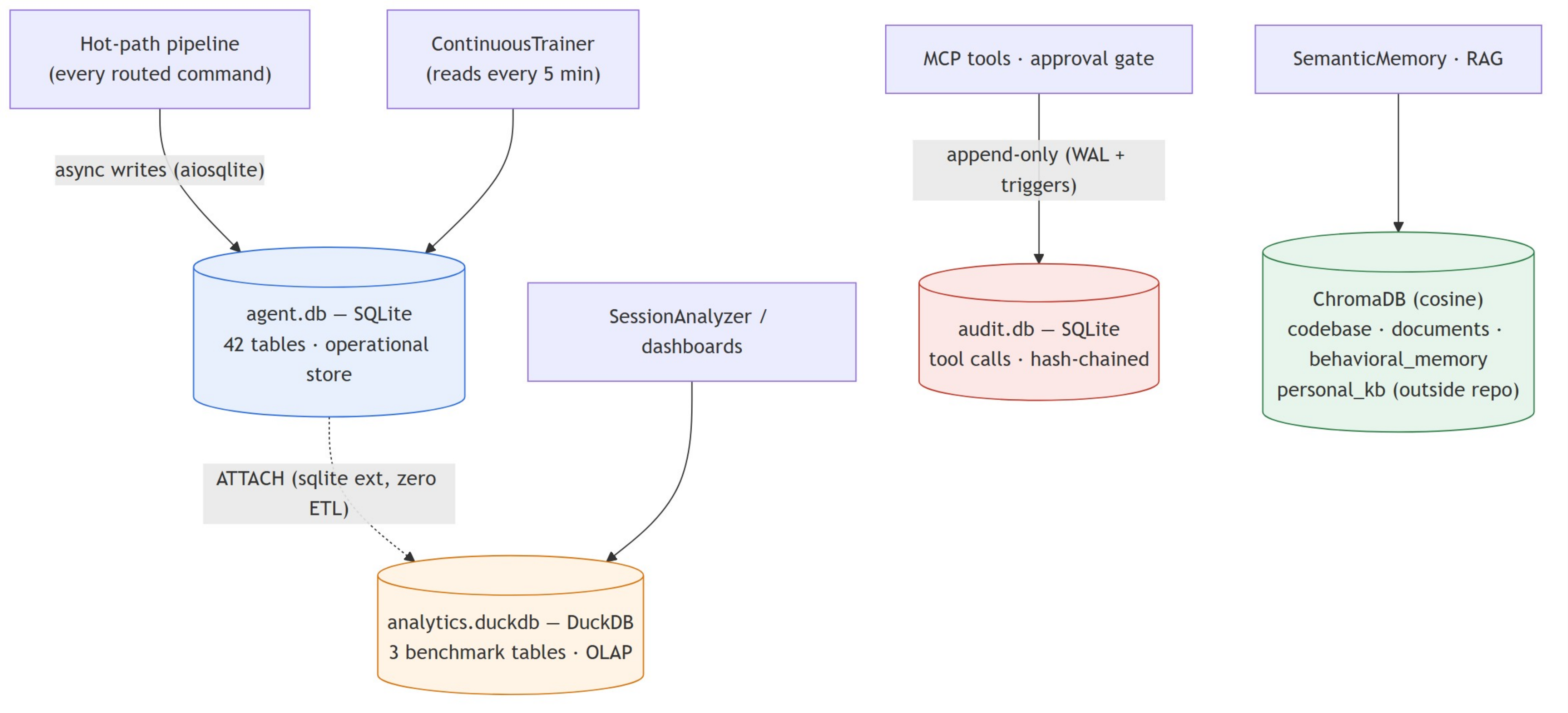
Data Model

Two-tier storage and the 42-table operational schema

Two-Tier Storage

Data Model

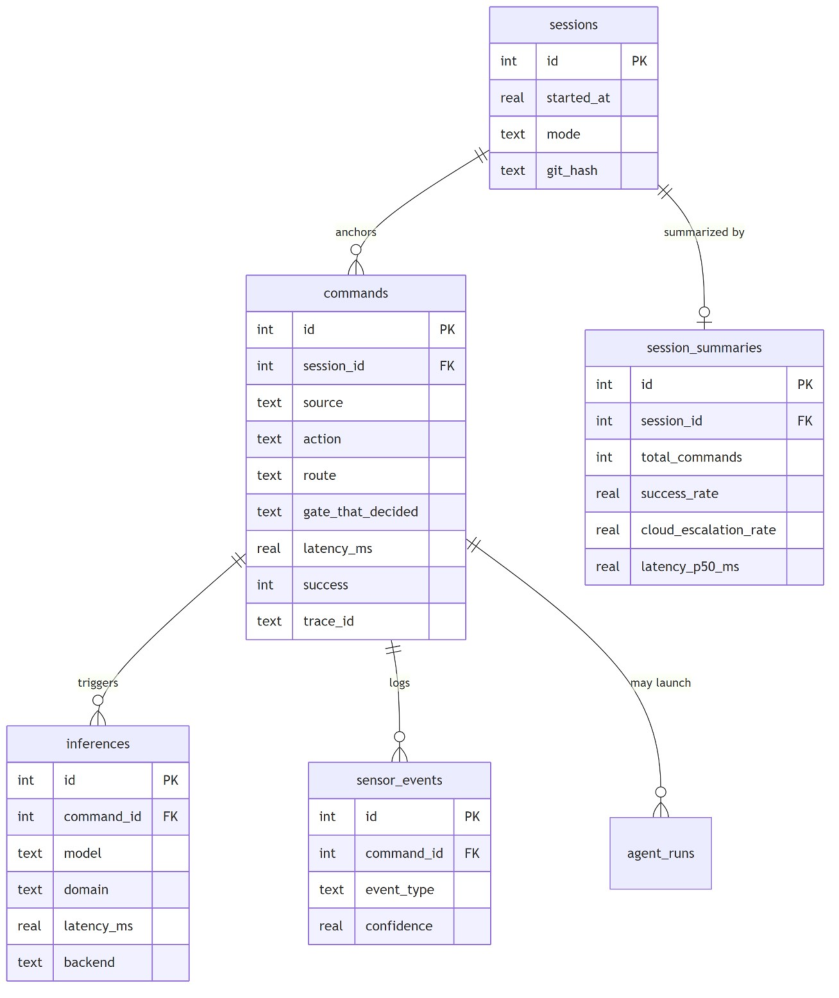
SQLite on the hot path, DuckDB for analytics (zero-ETL attach), ChromaDB for RAG, append-only audit.db



Core Star Schema

Data Model

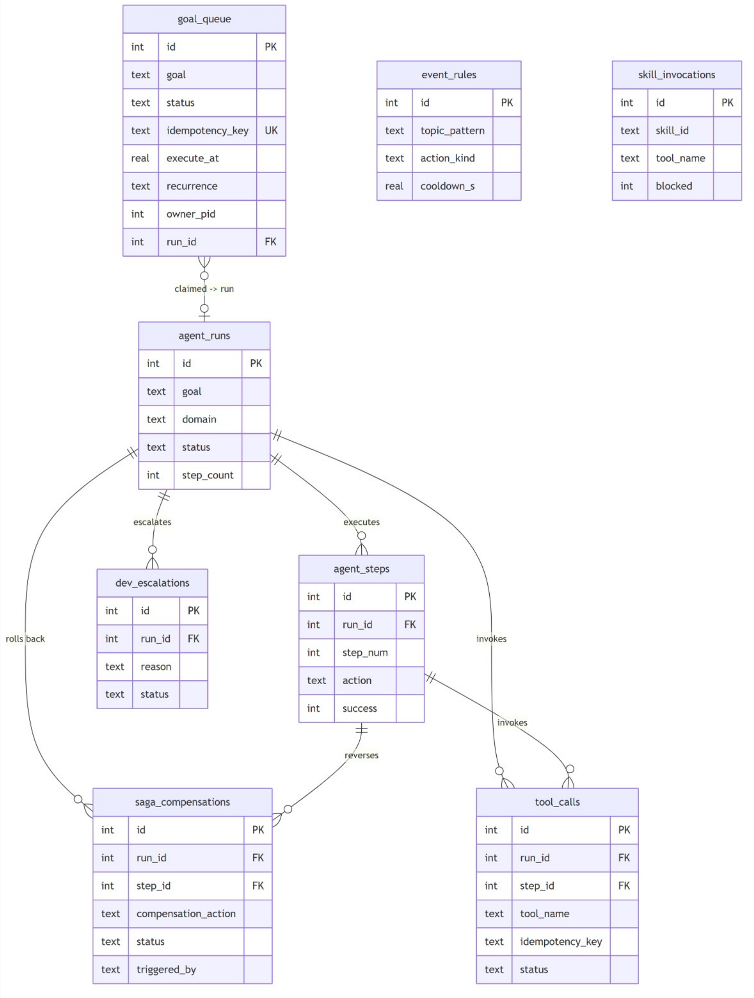
commands is the central fact table; sessions is the anchor



Dev-Agent Orchestration

Data Model

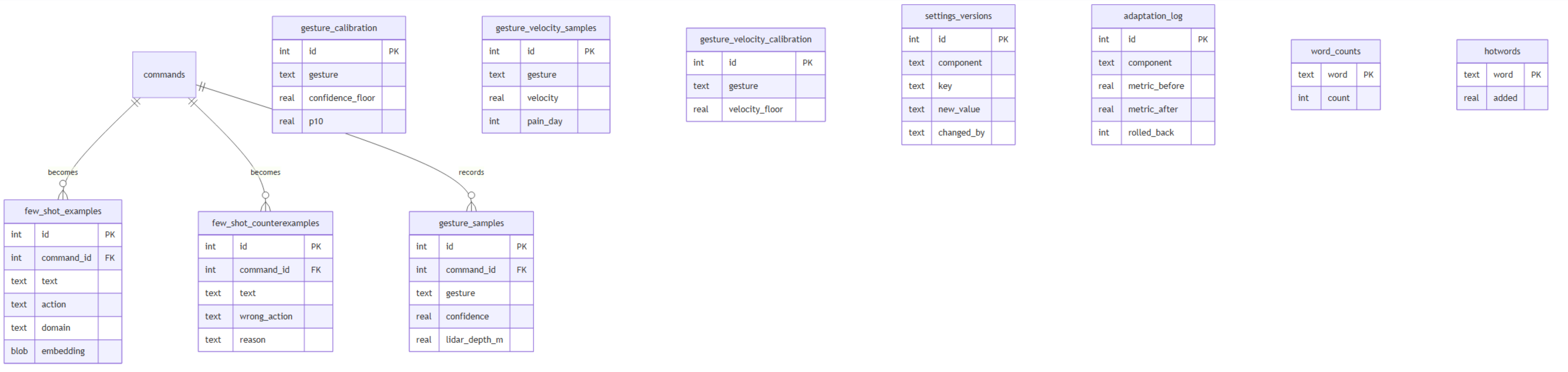
Durable goal queue → runs → steps, with saga rollback + escalation



Learning & Adaptation

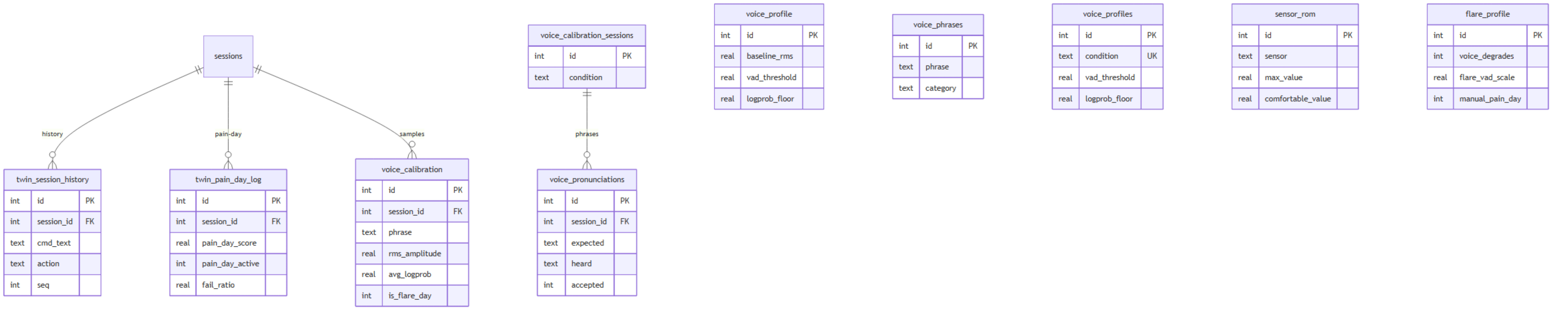
Data Model

Few-shot examples/counterexamples, gesture calibration, threshold history



Behavioral Twin & Voice

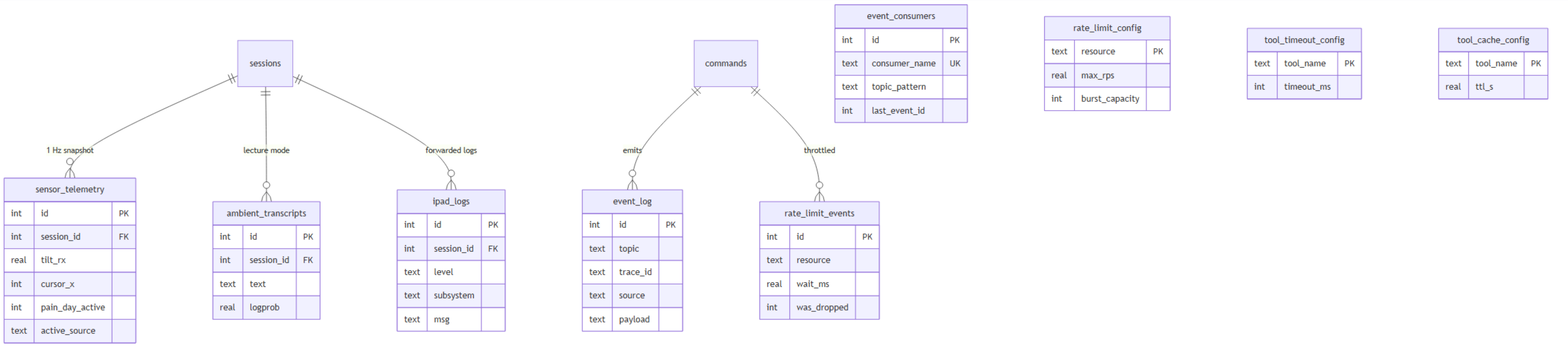
Pain-day signals, session history, and per-condition voice calibration



Telemetry, Events & Limits

Data Model

1 Hz sensor snapshots, the event bus, and rate-limit/observability config



Where Everything Lives

Diagrams (Mermaid source + PNG): docs/diagrams/{state,overview,db}/

State-machine spec: .kiro/specs/ipad-sensor-focus/diagrams/04-state-machines.md

DB design rationale: docs/architecture/database-design.md

Project orientation: CLAUDE.md (status, key files, conventions)

Rebuild this deck: python docs/diagrams/build_deck.py